gpeyt1-41 · Square footing, WT at base

GEO PE · Long Calculation Problems I — Topic 9: Shallow Foundations

Problem

A 6 ft × 6 ft column footing bears 4 ft below grade in a uniform medium-dense sand (c = 0, φ = 32°, γ = 120 pcf above and γsat ≈ 125 pcf below the water table). The seasonal-high water table is at the footing base (Dw = Df). Use the AASHTO/NHI general bearing equation with handbook shape factors and groundwater correction factors, FS = 3.0. Column dead load 160 kips, live load 90 kips (not needed to compute capacity).

Most nearly, the allowable column load (gross, FS = 3) is:

A. 295 kips
B. 256 kips
C. 199 kips
D. 148 kips

Given equations / data

> qult = cNc sc + qNq sq Cwq + 0.5γBNγ sγ Cwγ; φ = 32°: Nq = 23.2, Nγ = 30.2; sq = 1 + (B/L) tan φ; sγ = 1 − 0.4(B/L); Dw = Df → Cwq = 1.0, Cwγ = 0.5

Worked solution

1. q = 120×4 = 480 psf; sq = 1 + tan32 = 1.625; sγ = 0.6
2. Surcharge term = 480×23.2×1.625×1.0 = 18,095 psf
3. γ term = 0.5×120×6×30.2×0.6×0.5 = 3,262 psf
4. qult = 21,356 psf; qall = 7,119 psf
5. Qall = 7,119 × 36 ft² = 256 kips

Final answer

256 kips

Why the other values are wrong

- **295 kips** — ignored the groundwater correction (Cwγ = 1)
- **199 kips** — omitted shape factors
- **148 kips** — also took Cwq = 0.5 (misread the Dw = Df row)

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.4.2–3.4.2.1 Bearing Capacity Equation, Shape Factors, Bearing Capacity Factors and Groundwater Correction tables, pp. 90–91

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=du7B1MO-GCg, https://www.youtube.com/watch?v=ZUGdBao3tkc, https://www.youtube.com/watch?v=1fx03dLbmQc (no creator values copied).

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